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TIME AND WORK

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Introduction

Time and work are a fundamental arithmetic concept used to determine the relationship between the time taken to finish a task and the efficiency of the person or machine performing it. It is based on the idea that the total work done is the result of the rate of work multiplied by the total time spent

Sub - Categories in Time & Work:

1. **Time & Work with efficiencies**
2. **Chain rule**
3. **Work equivalence & Division of Wages**
4. **Pipes & Cistern**

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Concept and Definitions

Given below are a few such important time and work formulas for your reference:

Work Done = Time Taken × Rate of Work

Rate of Work = 1 / Time Taken

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If a piece of work is done in x number of days, then the work done in one day = 1/x

Total Wok Done = Number of Days × Efficiency

Efficiency and Time are inversely proportional to each other

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Key Formulae

1) If a pipe requires 'x' hours to fill up the tank, then part filled in 1 hr = 1/x
2) If a pipe requires 'y' hours to empty the full tank, then part emptied in 1 hr =1/y
3) Net Work Done = (Sum of work done by inlets) - (Sum of work done by outlets)
4) Suppose that one pipe takes 'x' hours to fill up the tank and another pipe takes 'y' hours to empty the full tank, then, on opening both the pipes, there are 2 possible conditions:

For condition I: $x < y$, Net part filled in 1 hr = $1/x - 1/y$

For condition II: $x > y$, Net part emptied in 1 hr = $1/y - 1/x$

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Questions

Ques.1 A can do a piece of work in 6 days and B can do the same work in 12 days. In how many days can they finish the work together?

a) 4 days

b) 3 days

c) 5 days

d) None of these

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Questions

Ques.2 Ajay and Vijay together can do a piece of work in 6 days. If Ajay alone, does it in 10 days, in how many days can Vijay alone do the work?

a) 12 days

b) 15 days

c) 18 days

d) None of these

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Questions

Ques.3 A is twice as good a workman as B. If B can finish a task in 30 days, in how many days can they complete the work together?

a) 12 days

b) 7 days

c) 10 days

d) None of these

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Questions

Ques.4 A is 60 percent more efficient than B. If A alone can do a piece of work in 25 days, in how many days can B alone do the same work (in days)?

a) 40

b) 60

c) 45

d) None of these

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Questions

Ques.5 A can do a piece of work in 15 days and the ratio of efficiency of A and B is 4 to 1. If they together earn 5400 rupees for the total work, what is B's wage?

a) 1080 rupees

b) 1350 rupees

c) 2700 rupees

d) 4320 rupees

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Questions

Ques.6 A is 30% more efficient than B. How much time will they take together to complete a job that A alone could have done in 23 days?

a) 21

b) 70

c) 49

d) 13

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Questions

Ques.7 A, B, and C can do a piece of work in 20, 30, and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

a) 15

b) 20

c) 25

d) None of these

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Questions

Ques.8 A is 50% as efficient as B. C does half of the work done by A and B together. If C alone does the work in 40 days, then in how many days can A, B, and C together do the work?

a)12

b) 13 (1/3)

c) 20

d) None of these

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Questions

Ques.9 20 workers can build a wall in 15 days. How many more workers are needed to complete the same wall in 10 days?

a) 10

b) 40

c) 5

d) None of these

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Questions

Ques.10 A and B undertook to do a piece of work for 600 rupees. A alone can do it in 6 days while B alone can do it in 8 days. With the help of C, they finish it in 3 days. What is the share of C?

a) 75 rupees

b) 100 rupees

c) 125 rupees

d) None of these

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Questions

Ques11. 6 men can do a piece of work in 12 days. 8 women can do the same work in 18 days and 18 children can do it in 10 days. 4 men, 12 women and 20 children work together for 2 days. If only men were to finish the remaining work in 1 day, how many total men are required?

a) 24

b) 36

c) 30

d) None of these

Questions

Ques12. Wages of 20 women for 15 days is 9000 rupees. If the daily wage of a man is one and a half times that of a woman, how many men must work for 30 days to earn 13500 rupees?

a) 10 men

b) 12 men

c) 20 men

d) None of these

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Questions

Ques13. Two pipes A and B can fill a tank in 20 minutes and 30 minutes respectively. If both pipes are opened together, how much time will they take to fill the tank?

a) 10 minutes

b) 12 minutes

c) 15 minutes

d) None of these

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Questions

Ques14. Two pipes can fill a tank in 10 hours and 12 hours respectively while a third pipe empties the full tank in 20 hours. If all the three pipes operate together, in how much time will the tank be filled?

a) 7 hrs

b) 8 hrs

c) 7.5 hrs

d) None of these

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Questions

Ques15. Two pipes A and B can fill a cistern in 12 minutes and 15 minutes respectively, but a third pipe C can empty the full tank in 6 minutes. A and B are kept open for 5 minutes and then C is also opened. In what time is the cistern emptied?

a) 30 minutes

b) 40 minutes

c) 45 minutes

d) None of these

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Thank You